

DocScribe

Multilingual OCR Pipeline

Hello

こんにちは

नमसे

Hola



Bonjour

你好

मरति



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THE PROBLEM

Documents need to have their textual content extracted reliably.

Extracted text must be consumable by downstream systems such as:

- 🔍 Search indexing
- 🤖 Data-entry automation
- 🛡️ Compliance checks
- 📊 Analytics

The absence of a robust, language-aware OCR pipeline creates challenges in processing:

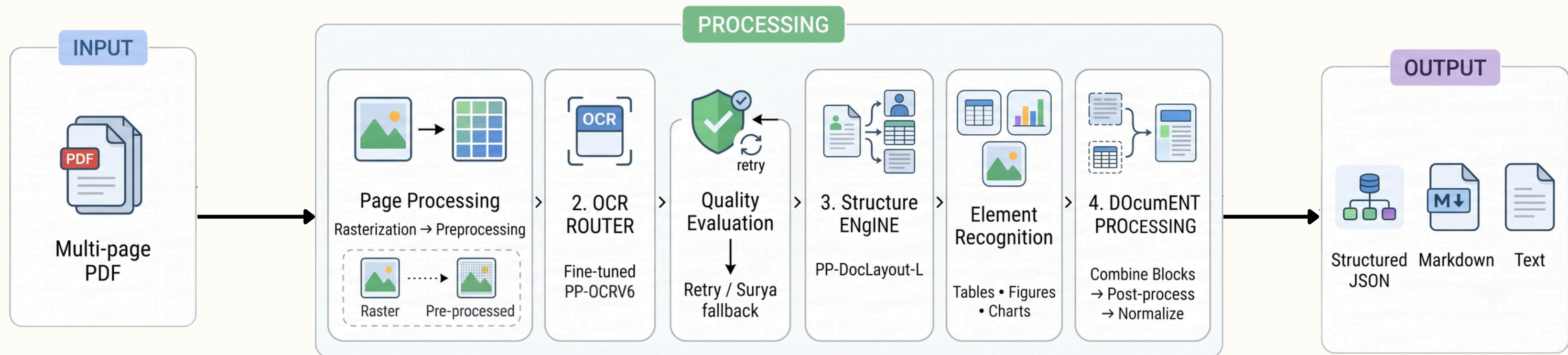
- Multilingual documents
- Structurally complex documents
- These documents may fail to process or produce low-quality, unreliable text output



THE SOLUTION – DOCSCRIBE

DocScribe is an end-to-end open-source OCR pipeline designed to:

An end-to-end, open-source OCR pipeline that accepts multi-page PDF documents as input and produces clean, structured, and language-correct extracted text as output, suitable for direct use by downstream systems without manual correction.



PDF Upload → Page/Image → Preprocess → Fine Tuned - Traditional OCR → Confidence Check → VLM based OCR → Normalize → JSON/Markdown

FEATURES



Multilingual OCR

4-Language Recognition

English, Hindi, Marathi & Gujarati with fine-tuned recognition weights



Structured Extraction

Text + Tables + Figures —

bounding boxes, confidence, language & structured cell data



Adaptive Retry

Quality-Aware Retry

enhanced / threshold variants before secondary-engine fallback



OCR Quality Scoring

5-Signal Quality Score

drives retry and fallback decisions, not just reporting



Optional AI Correction

Controlled Post-OCR

Correction API-based, safety-constrained and disabled by default

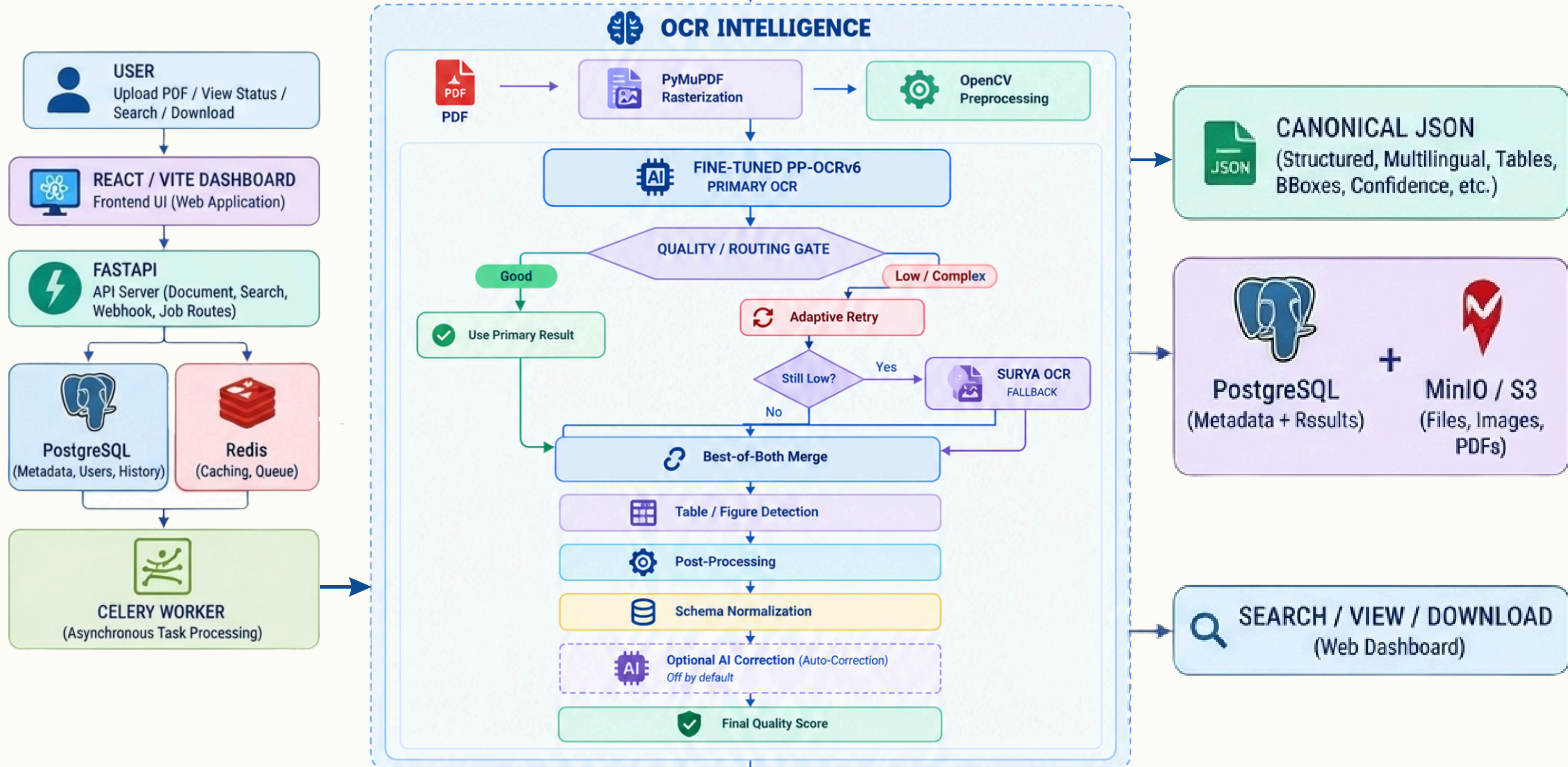


Multi-Format Output

6 Export Formats

JSON, Markdown, TXT, Searchable PDF, Highlighted PDF & Structured PDF

SYSTEM ARCHITECTURE



HYBRID OCR STRATEGY

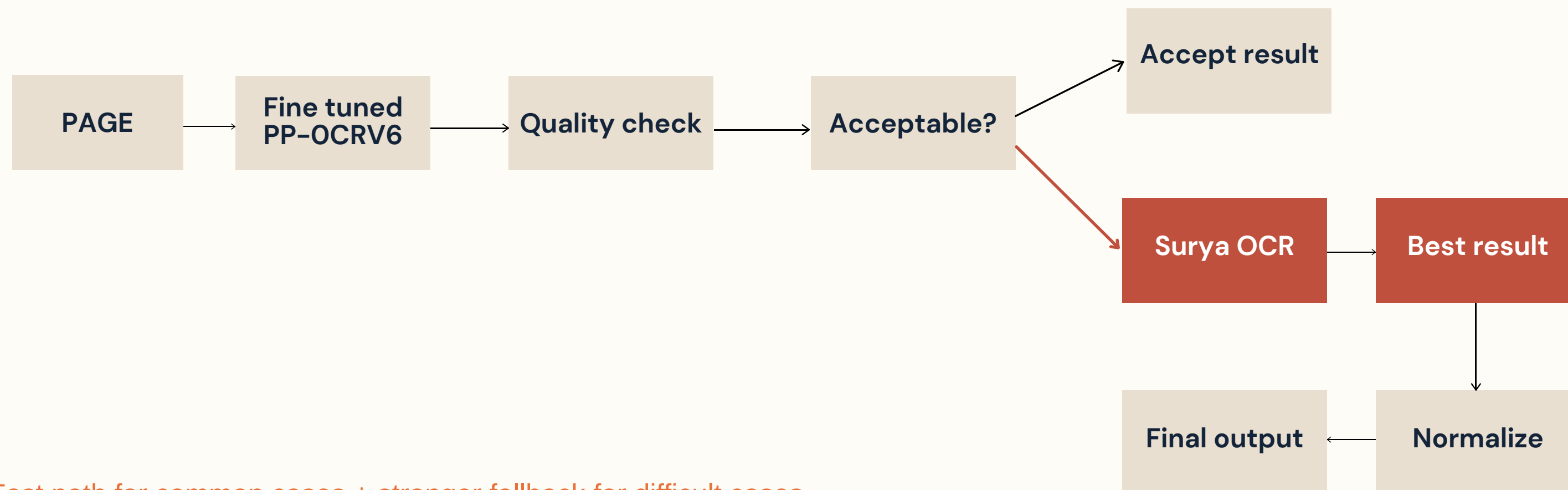
“Why Hybrid OCR?”

Traditional-based **Primary Engine** Fine Tuned PP-OCRv6

- Fast
- Suitable for normal pages
- CPU-friendly
- Handles the majority of pages

VLM-based **Fallback Engine** Surya OCR

- Used for difficult pages
- Useful for complex layouts
- Used when primary OCR results are not reliable
- More computationally expensive



Fast path for common cases + stronger fallback for difficult cases.

FINE-TUNED PP-OCRV6 RECOGNITION

Custom fine-tuning of the PP-OCRv6 recognition component for improved multilingual document text recognition.

Version	Training Setup	Languages	Dictionary / Characters	Epochs	Batch Size	Learning Rate	Result / Status
V1	Default pretrained model	8	~38K chars	10	6	0.0005	90.26% accuracy ; incorrect predictions observed; Kannada relatively strong, Bengali
V2	Fine-tuned on custom dataset	8	~171K chars	15	6	0.0005	98.25% accuracy , ~99.50% normalized edit similarity; occasional incorrect predictions remained
V3	Fine-tuned on default/pretrained model	8	~70K dictionary chars	20	8	0.0003	Training halted/cancelled ; not used as final model
V4	Fresh fine-tuning from original PP-OCRv6 pretrained checkpoint	4	326 dictionary chars; 287 used; 0 missing	20	8	0.0003	60.18% validation accuracy , 97.29% edit similarity; final 4-language experiment
V5	Final Model	4	660-character dictionary	20	8	—	Final model used ~57.25% accuracy, ~92.31% normalized edit score; 23 fonts, 2 workers

Detection architecture remains PP-OCRv6; our contribution is the fine-tuned recognition weights integrated into the production pipeline.

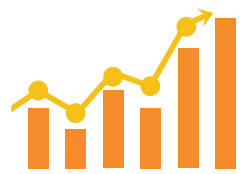
TECH STACK

Layer	Technology	Why It Is Used
Frontend	React + Vite + Tailwind CSS	Build an interactive interface for PDF upload, job status, and OCR results
Backend	FastAPI	Provide the API and connect the frontend with the Python-based processing pipeline
Job Queue	Celery + Redis	Run OCR tasks asynchronously without blocking the API
Primary OCR	PaddleOCR	Provide a fast OCR path for normal document pages
Fallback OCR	Surya OCR	Handle difficult pages where the primary OCR result is unreliable
PDF Processing	PyMuPDF	Convert PDF pages into images for OCR processing
Preprocessing	OpenCV	Improve scanned-page quality through deskewing, denoising, and binarization
Language ID	PaddleOCR + Language-specific models	Identify the document/page language for language-aware processing
Database	PostgreSQL	Store structured data such as documents, jobs, pages, and OCR metadata
Object Storage	MinIO/ S3	Store uploaded PDFs and generated processing outputs
Deployment	Docker Compose	Run and manage the project's multiple services in a consistent environment

OUR USP



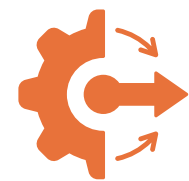
Fine-Tuned Recognition - Custom PP-OCRv6 recognition weights optimized for 4 languages



Quality-Aware Routing - OCR quality actively determines retry and fallback decisions



Multi-Engine Resilience - Difficult pages can move from primary OCR → retries → secondary OCR



Structured-First Architecture - A canonical structured representation drives all downstream outputs



Traceable Processing - OCR attempts, confidence, quality and corrections remain auditable

CONCLUSION

DocScribe is a **deployable, open-source multilingual OCR pipeline** that combines **PaddleOCR** with confidence-based **Surya** fallback routing to process complex, multi-page documents efficiently.

It supports **English, Hindi, Marathi** and **Gujarati**, with **structured JSON/Markdown** output, **table** handling, **chart** safety, and scalable processing.

DocScribe is an **OCR backbone** for search indexing, data-entry automation, and compliance workflows

Expected impact: reduces the manual-correction bottleneck for multilingual document processing — with the potential to reduce dependency on paid cloud OCR APIs.

"DocScribe turns multilingual OCR from a manual bottleneck into a scalable, structured pipeline."

THANK YOU !